Case Study 4: ATM Transaction Sequence Simulation (PDF Case 13)

1. Problem Statement

Simulate a sequence of deposits and withdrawals, update the account balance, reject invalid transactions, and summarize successful transactions.

2. Objectives

Use loops and conditional statements.

Update a running balance.

Count successful deposits and withdrawals.

3. Assumptions

The opening balance cannot be negative.

Transactions are entered as D for deposit or W for withdrawal.

Amounts must be positive.

A withdrawal is rejected if it exceeds the available balance.

4. Algorithm

1. Start.

2. Input opening balance and number of transactions.

3. Repeat for each transaction: input type and amount.

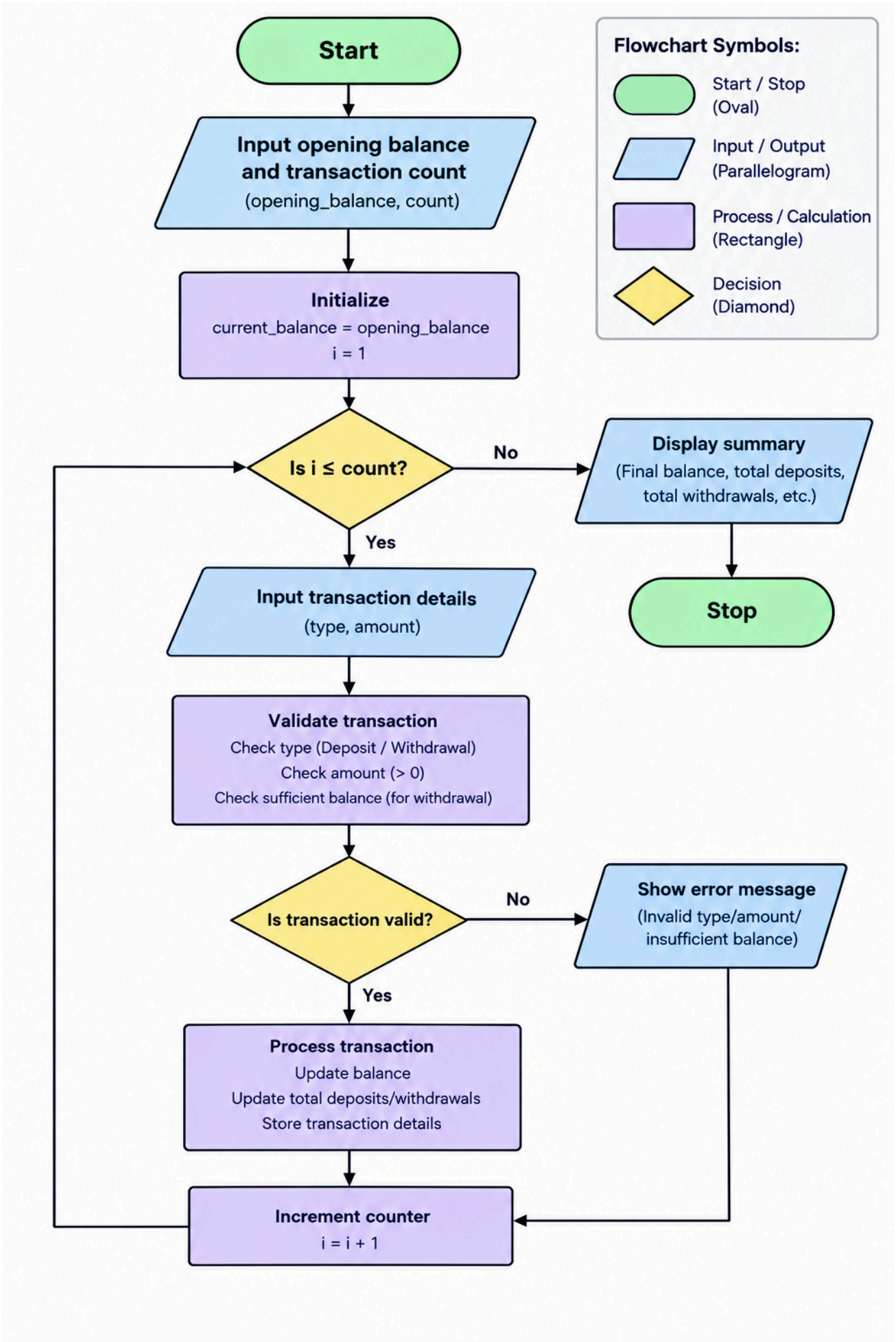
4. Accept valid deposits; accept withdrawals only when sufficient balance exists.

5. Count successful deposits and withdrawals.

6. Display final balance and transaction counts.

7. Stop.

5. Flowchart (text form)


Start
Input opening balance and transaction count
(opening_balance, count)
Initialize
current_balance = opening_balance
i = 1
Is i ≤ count?
No
Display summary
(Final balance, total deposits, total withdrawals, etc.)
Stop
Yes
Input transaction details
(type, amount)
Validate transaction
Check type (Deposit / Withdrawal)
Check amount (> 0)
Check sufficient balance (for withdrawal)
Is transaction valid?
No
Show error message
(Invalid type/amount/ insufficient balance)
Yes
Process transaction
Update balance
Update total deposits/withdrawals
Store transaction details
Increment counter
i = i + 1
Flowchart Symbols:
Start / Stop (Oval)
Input / Output (Parallelogram)
Process / Calculation (Rectangle)
Decision (Diamond)

6. Pseudocode

START
INPUT balance, n
SET deposits = 0, withdrawals = 0
REPEAT n times: INPUT type, amount
IF amount <= 0: DISPLAY invalid amount
ELSE IF type is D: balance = balance + amount; deposits = deposits + 1
ELSE IF type is W AND amount <= balance: balance = balance - amount; withdrawals = withdrawals + 1 ELSE DISPLAY transaction rejected
DISPLAY balance and counts
STOP

7. Complete Python Source Code

```
balance = float(input("Enter opening balance: Rs. "))
```



```
while balance < 0:
 print("Balance cannot be negative.")
 balance = float(input("Enter opening balance: Rs. "))

n = int(input("How many transactions? "))
deposits = 0
withdrawals = 0
for i in range(n):
kind = input("Enter transaction type (D for deposit, W for withdrawal):
").strip().upper() amount = float(input("Enter amount: Rs. "))
 if amount <= 0:
 print("Transaction rejected: amount must be positive.")
elif kind == "D":
balance += amount
deposits += 1
 print("Deposit successful.")
elif kind == "W":
 if amount <= balance:
balance -= amount
withdrawals += 1
 print("Withdrawal successful.")
 else:
 print("Transaction rejected: insufficient balance.")
 else:
 print("Transaction rejected: invalid transaction type.")

print("
----- ATM SUMMARY -----")
print(f"Final balance: Rs. {balance:.2f}")
print("Successful deposits:", deposits)
print("Successful withdrawals:", withdrawals)
```

8. Test Case and Expected Output

Inputs: opening balance 1000; 3 transactions: D 500, W 200, W 2000.

Deposit successful. Withdrawal successful. Transaction rejected: insufficient balance. Final balance: Rs. 1300.00
Successful deposits: 1 Successful withdrawals: 1

9. Conclusion

The program simulates ATM transactions while checking transaction type, amount, and available balance.